

Engineering Chemistry (CHM1PF01A)
Unit 1: Specialty Polymers for Engineering Applications
 (Lecture Note 1)

1. Monomer

- (1) Monomers are the building blocks of polymers. It forms linkages with other monomer molecules to form the macromolecule.
- (2) The word 'monomer' is derived from two Greek words namely,
 - (a) 'Mono' means single and
 - (b) 'Meros' means parts
- (3) The functional groups in the monomer organic molecules can be like :
 - OH, – COOH, – NH₂, – C ≡ N, – COOR, – Cl, cyclic amide (lactam) group, cyclic acid anhydride group, etc.

A C = C contains a weak π -bond and it can easily be broken by the action of heat or a reagent. When the π -bond is broken, two positions become available for polymerization.

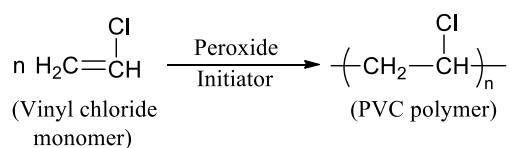
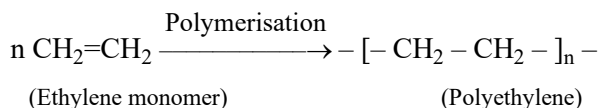
2. Polymer

- (1) **Definition:** A polymer is a large molecule built by the repetition of small and simple chemical unit called monomer. The repeating chemical units are covalently linked to each other in a macromolecule.
- (2) The word 'polymer' is derived from two Greek words namely,
 - (a) 'Poly' means many and
 - (b) 'Meros' means parts.

3. General characteristics of polymers

- (1) Polymeric molecules are very big molecules. Their average molecular weights may approach 10^5 or more.
- (2) Polymers are semi-crystalline materials. It means they have both amorphous and crystalline regions.
- (3) The intermolecular forces in polymers can be vander waals forces, dipole-dipole.
- (4) Hydrogen bonding attraction.
- (5) Polymers are combustible materials.
- (6) Generally, polymers are thermal and electrical insulators. Although, number of conducting polymers have been discovered so far.

4. Polymerization reaction: The process with the help of which monomers are transformed into a polymer is called polymerization reaction.



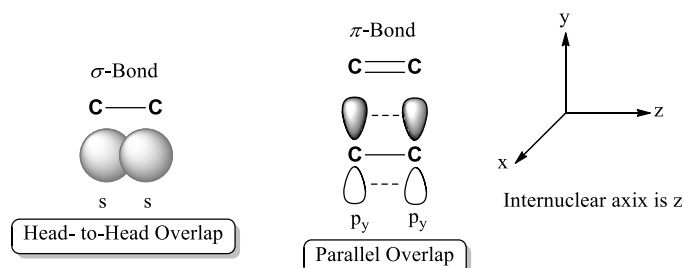
5. Functionality:

Functionality is defined as the number of bonding sites available in a single monomer unit.

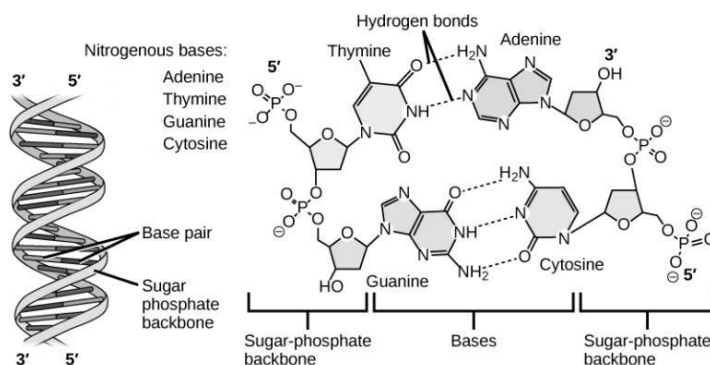
Monomer Unit	Functionality
$\text{CH}_2=\text{CH}_2$ (Ethylene)	2
$\begin{array}{c} \text{H} \\ \\ \text{H}_3\text{C}-\text{C}=\text{CH}_2 \end{array}$ (Propene)	2
$\begin{array}{c} \text{H}_2\text{C}-\text{OH} \\ \\ \text{H}_2\text{C}-\text{OH} \end{array}$ (Ethylene Glycol)	2
$\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{NH}_2$ (Hexamethylene-diamine)	2
$\text{HOOC}-\text{C}_6\text{H}_4-\text{COOH}$ (Terephthalic acid)	2

6. Chemical Bonding in Polymer

- (a) **Covalent Bond:** Its an interatomic linkage formed due to equal sharing of electron pair between two atoms. The most prevalent C-C σ and π -bonds are usually found in polymer molecules. Carbon has higher catenation property which is due to the strength of its covalent bonds that allows two or more carbon atoms to share their valence electrons.



- (b) **Hydrogen Bond (Van der Waals bond):** It is a type of dipole-dipole attraction between H and electronegative atoms. Hydrogen bond can be intermolecular and intramolecular. H-bonding in DNA polymer is mentioned below.



7. Molecular weight of polymers

(a) Number-Average Molecular mass $\overline{M}_n$

Number-average molecular mass ($\overline{M}_n$) is defined as the total mass (M) of all the molecules in a polymer sample divided by the total number of molecules present in the sample.

$$\overline{M}_n = \frac{\sum N_i M_i}{\sum N_i} = \frac{N_1 M_1 + N_2 M_2 + \dots}{N_1 + N_2 + \dots}$$

N = Number of molecule in polymer chain.

M = Molecular weight of the fraction.

(b) Weight-Average Molecular-Mass $\overline{M}_w$

Weight-average molecular mass ($\overline{M}_w$) is defined as the total weight (W) of all the molecules in a polymer sample divided by the weight-fraction of molecules present.

$$\begin{aligned} \overline{M}_w &= \frac{\sum W_i M_i}{\sum W_i} \\ &= \frac{\sum N_i M_i^2}{\sum N_i M_i} \quad (\text{Where, } W = N \times M) \end{aligned}$$

(c) Viscosity-Average Molecular mass $\overline{M}_v$

The average molecular weight of polymer is obtained from viscosity measurement is known as viscosity average molecular weight and it is expressed mathematically as follows;

$$\overline{M}_v = \left[\frac{\sum W_i M_i^\alpha}{\sum W_i} \right]^{1/\alpha} = \left[\frac{\sum N_i M_i^{1+\alpha}}{\sum N_i M_i} \right]^{1/\alpha}$$

α is the exponent of Mark-Hawkin-Sakurada equation. When $\alpha = 1$, $\overline{M}_v = \overline{M}_w$

Generally, value of α is $0.5 < \alpha < 0.9$. The value of α depends upon nature of polymer, solvent and temperature.

8. Polydispersity index (PDI)

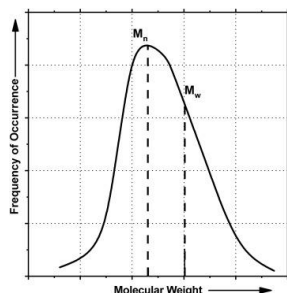
PDI is defined as the ratio of $\overline{M}_w$ and $\overline{M}_n$.

$$\text{PDI} = \frac{\overline{M}_w}{\overline{M}_n} \quad \text{PDI} = 1 \text{ for monodisperse polymer.}$$

$\text{PDI} > 1$ for polydisperse polymer

9. Molecular weight distribution curve

Mixture of many systems present in polymer which have different molecular weights. In order to characterize polymers, we need use the molecular weight distribution (MWD) curve, which represents a plot of the frequency of occurrence for a particular species against its molecular weight. As a result of the existence of different sizes of molecular species in a polymeric material, we cannot strictly assign of the molecular weight of a polymer. Instead, we use molecular weight averages to express the size of synthetic polymers.



Molecular Weight	Methods of Determination	Molecular Weight	Methods of Determination
$\overline{M}_n$	End Group Analysis	$\overline{M}_w$	Light Scattering
	Ebullioscopy		Sedimentation
	Cryoscopy	$\overline{M}_v$	Viscometry
	Osmometry		
	Gel Permeation Chromatography (GPC)		

10. Numerical problems on molecular weights

P1: A polymer sample consists of 10 chains with the following molecular weights calculate number-average and weight average molecular weight and PDI of the sample.

Number of Molecules (N_i)	2	3	5
Molecular weight of each molecule (M_i)	50000	15000	10000

We know that,

$$(I) \quad \overline{M}_n = \frac{\sum N_i M_i}{\sum N_i} = \frac{2 \times 50000 + 3 \times 15000 + 5 \times 10000}{10} = \mathbf{19500}$$

$$(II) \quad \sum N_i M_i = 2 \times 50000 + 3 \times 15000 + 5 \times 10000 = \mathbf{195000}$$

$$(III) \quad \overline{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i} = \frac{2 \times (50000)^2 + 3 \times (15000)^2 + 5 \times (10000)^2}{195000} = \mathbf{31666.666}$$

$$(IV) \quad PDI = \frac{\overline{M}_w}{\overline{M}_n} = \frac{31666.666}{19500} = \mathbf{1.62}$$

Hence, the polymer is polydisperse.

(Some practice problems)

Q. 1 In a sample of polymer, there are 3 molecules. 50% molecules have molecular weight of 60000, 20% molecules have molecular weight of 20000 and 30% molecules have molecular weight of 40000. Calculate number average and weight average molecular weight of the sample.

(Ans. : $\overline{M}_n = 43000$, $\overline{M}_w = 50000$)

Q. 2 A polymer has molecules having molecular weights of 2×10^5 , 3×10^5 and 1×10^5 . Calculate the number-average molecular weight of the polymer. (Ans. : **200000**)

- Q. 3** A polymer sample has 40%, 30%, 20% and 10% molecules of molecular weights of 16000, 14000, 12000 and 10000 respectively. Find number-average molecular weight as well as weight-average molecular weight of the polymer sample. **(Ans. : 14000, 14300)**
- Q. 4** A polymer with 10 chains has 3 molecules of molecular weight of 30000, 2 molecules of molecular weight of 60000 and 5 molecules of molecular weight of 10000. Calculate number-average and weight-average molecular weights of the given polymer. **(Ans. : $\overline{M}_n = 26000$, $\overline{M}_w = 39999$)**

11. Classification of polymers

On the Basis of	Types
Occurrence	(a) Natural polymers (b) Synthetic or organic polymers
Structure of chain	(a) Linear polymers (b) Branched polymers (c) Cross linked polymers
Stereo chemical orientation	(a) Isotactic (b) Syndotactic polymers (c) Atactic polymers
Repeating units	(a) Homopolymers (b) Co-polymers
Synthesis	(a) Addition polymers (b) Condensation polymers
Thermal behavior	(a) Thermoplastic polymers (b) Thermosetting polymers or thermosets

(I) On the basis of occurrence

(a) Natural polymers : Those polymers which are obtained from natural sources (plants and animals) are called natural polymers

Example : Starch, cellulose, rubber, DNA protein, silk etc.

(b) Synthetic or organic polymers: These polymers are synthesized by men. Most of the synthetic polymers are long chain organic molecules containing thousands of monomer units.

Example : Polythene, nylon, PVC etc.

(II) On the basis of chain structure

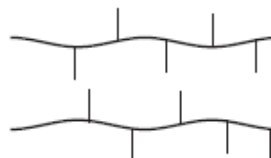
(a) Linear polymers : In linear polymers, each monomeric unit is linked with two other monomeric units on either side, forming a continuous straight chain.

The monomeric units are added on to each other forming a long chain.

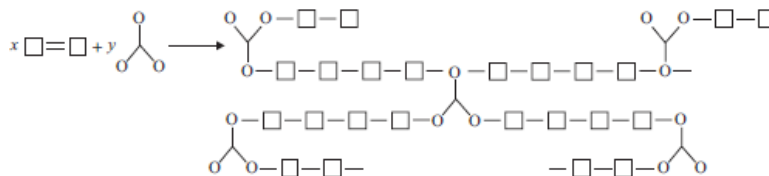
Example : High density polythene (HDPE) structure

(b) Branched polymers : In branched polymers most of the monomeric units are linked with two other on either side. Some monomeric units are linked with a third monomeric unit.

Example : Low density polythene (LDPE)



- (c) **Cross linked polymers:** When a bi-functional monomer is mixed in small amount with tri-functional monomers, a three dimension network polymer is formed which is known as cross-lined polymer.

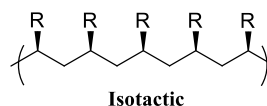


(III) On the basis of stereo chemical orientation

On the basis of tacticity, polymers are of following three types:

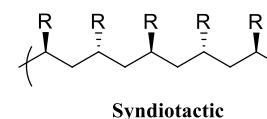
- (a) **Isotactic polymer :** In this type of polymer all the functional groups are present on the same side of the chain i.e. ordered orientation.

Example : Polypropylene



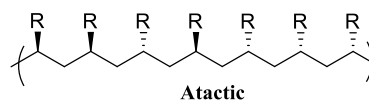
- (b) **Syndiotactic polymers :** If the arrangement of functional group is in alternating fashion around the main chain, it is called syndiotactic polymers.

Example : Natural rubber



- (c) **Atactic polymers:** In this type of polymer the arrangement of functional groups are at random around to the main chain.

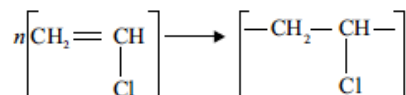
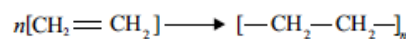
Example : Polystyrene



(IV) On the basis of repeating units

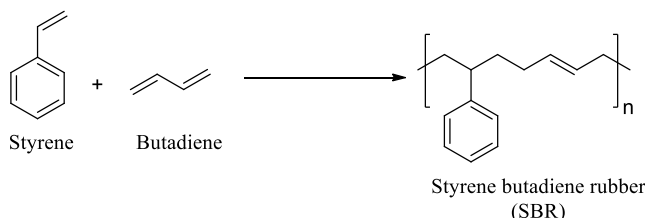
- (a) **Homopolymers:** Homopolymers are those polymers where the entire polymer chain is composed of one single repeat unit.

Example: Polyethylene, PVC



- (b) **Co-polymers :** Co-polymers are those polymer which have more than one type of repeating unit in the polymer backbone.

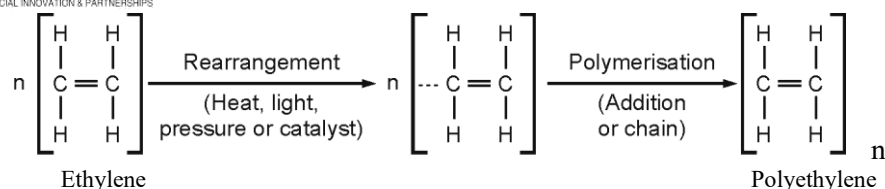
Example:



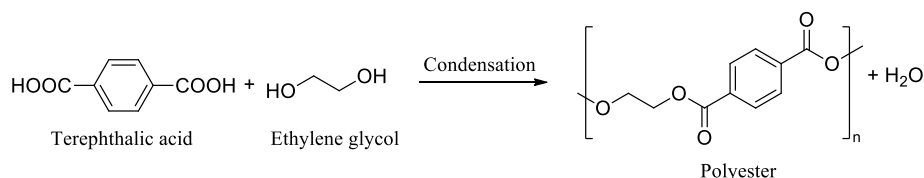
(V) On the basis of synthesis

- (a) **Addition polymer:** A polymer formed by direct addition of repeated units of monomer is called addition polymer.

E.g. Polyethylene, Orlon, PVC etc.



(b) Condensation polymers: A polymer formed by condensation of two same or different monomers with the liberation of by-product is known as condensation polymer. E.g. Polyester, Nylon 66, Kevlar, Polycarbonate etc.



(VI) On the basis of thermal behavior

(a) Thermoplastic polymers: These polymers are flexible, linear chain molecule which once shaped or formed can be softened by the application of heat and can be reshaped repeatedly.

Examples: Polyethylene, Nylon, Teflon, polycarbonate etc.

(b) Thermosetting polymers: These are rigid and 3D polymeric network, which once shaped or formed can't be softened by the application of heat.

Examples: Polyester, Bakelite, Thermoset resin, Melamine formaldehyde polymer etc.

S.No.	Thermoplastics	Thermosetting
1.	These are the products of addition polymerizations	Products of condensation polymerization
2.	Long chain linear polymers	Have three-dimensional network structure
3.	Monomer used in the production is bi-functional	Used higher functional monomers
4.	Low molecular weight	High molecular weight
5.	They are soft, weak and brittle	They are hard, strong and brittle
6.	Soluble in organic solvents	They are insoluble
7.	Softened on heating readily	They do not soften on heating
8.	They can be softened, reshaped and recycled	Retain their shape and structure